# Car Price Prediction

Milestone 2 — Data Preparation

## INTRODUCTION AND REPRODUCIBILITY

This document describes the complete data preparation pipeline for the Car Price Prediction project. The raw Kaggle dataset is transformed into a clean, fully engineered, and machine-learning-ready feature matrix. A fixed random seed (**RANDOM_SEED = 42**) is applied to every operation involving randomness, ensuring any collaborator who re-runs the notebook obtains identical results.

## IMPROVING DATA QUALITY

### Hidden Missing Values

The raw dataset uses a dash character (**"-"**) as a placeholder for missing entries rather than standard NaN — most prominently in the **Levy** (vehicle import tax) column, affecting approximately 30% of rows.

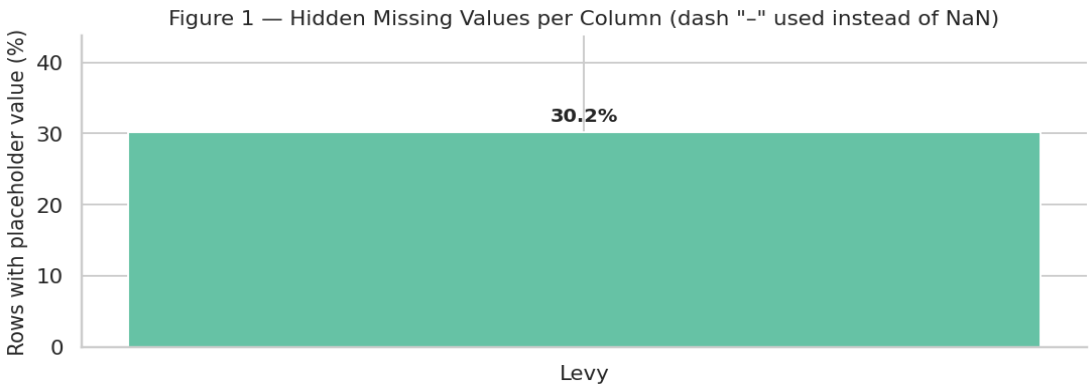


Figure 1 — Dash-placeholder missing values per column. Levy is most affected (~30%).

### Type Conversion

- **Mileage:** "186005 km" → 186005 (strip unit suffix, cast to integer)
- **Engine volume:** "2.0 Turbo" → 2.0 via regex; "Turbo" stored separately as a binary feature
- **Levy:** Remove thousands comma, cast to float
- **Price:** Cast to numeric; rows that fail are dropped — the target cannot be imputed

### Imputation

Remaining numeric NaN values are filled with the column **median**. The median is preferred over the mean for skewed columns (Levy, Mileage) where extreme values distort the mean. Rows with missing categorical values are dropped entirely.

## DISTRIBUTIONS

The price distribution is heavily right-skewed. The log scale reveals the full range and exposes the cluster of near-zero erroneous entries.

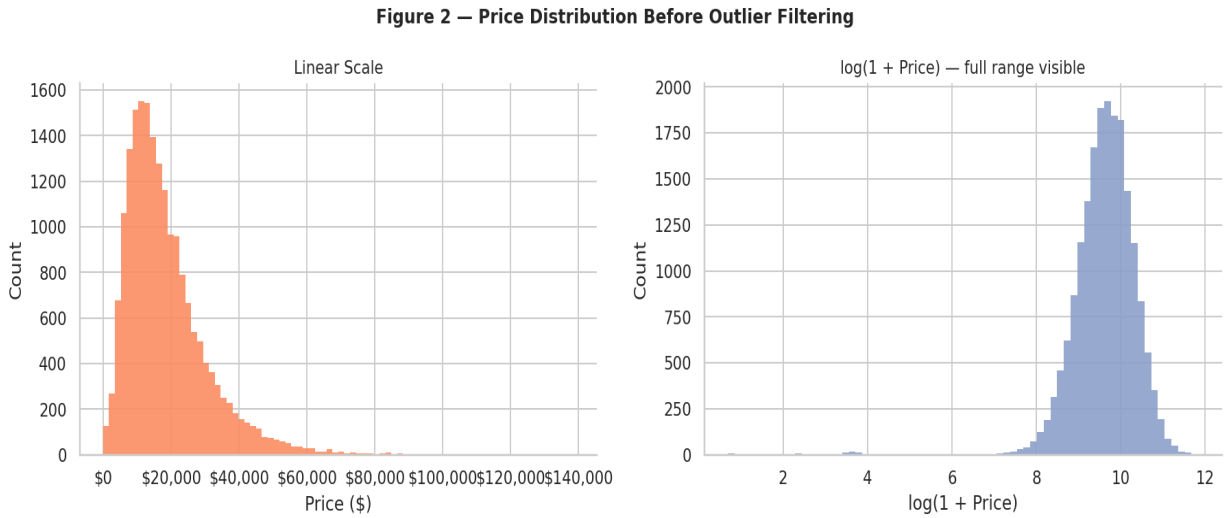


Figure 2 — Price distribution before filtering. Linear (left) shows skew; log scale (right) shows full range.

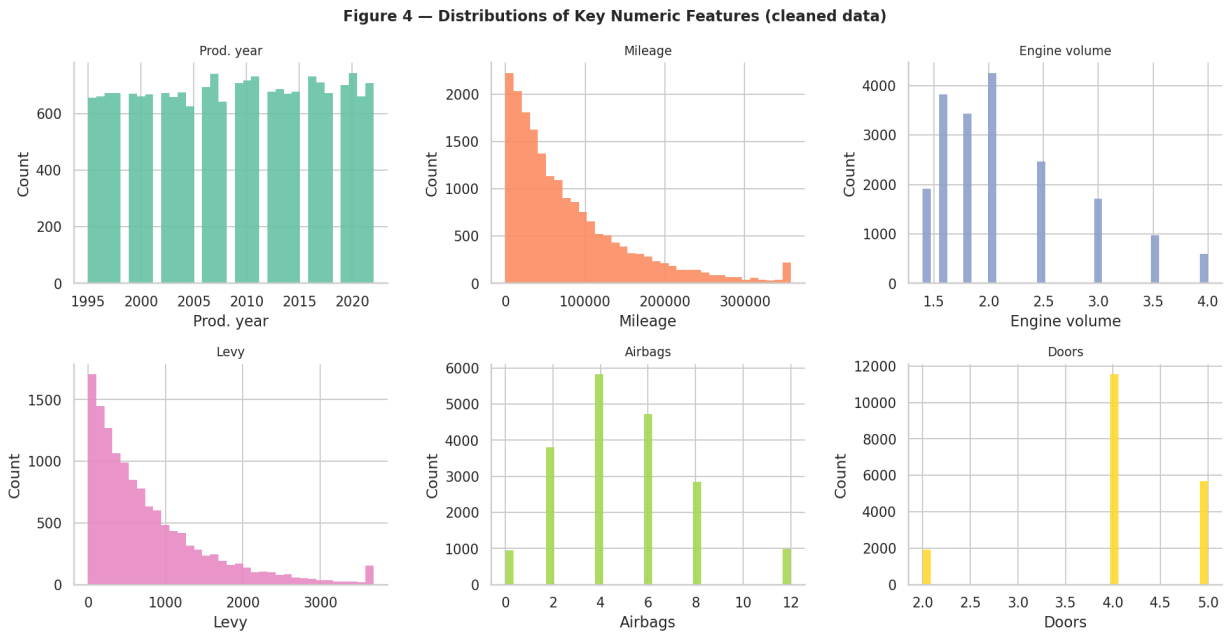


Figure 3 — Distributions of six key numeric features. Production year skews left; mileage and levy are right-skewed.

The correlation matrix confirms M1 findings: **Production Year** has the strongest positive correlation with Price; **Mileage** has the strongest negative.

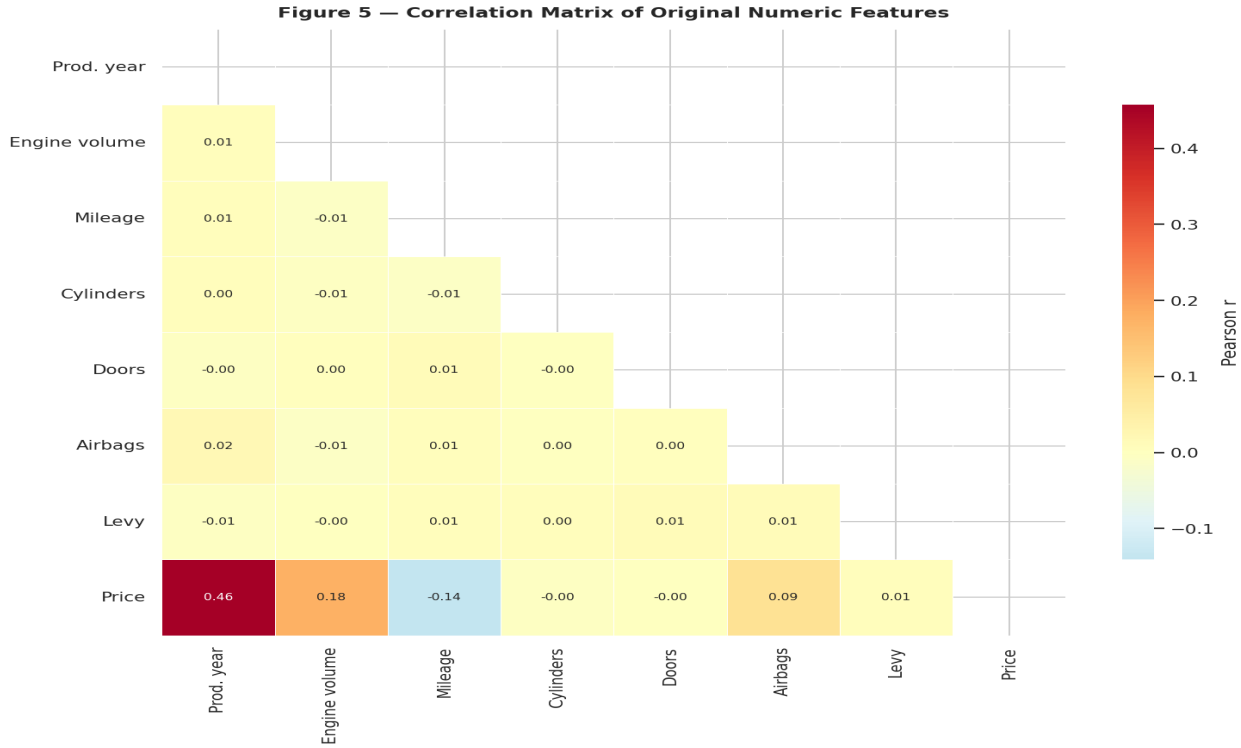


Figure 4 — Full correlation matrix of original numeric features.

## OUTLIER INVESTIGATION AND FILTERING

- **Lower bound ($500):** Prices of $1–$2 are data entry errors with no real-world meaning. All listings below $500 are removed.
- **No upper bound:** Luxury cars above $100,000 are valid real-world listings. Removing them would bias the model toward underpredicting premium vehicles.

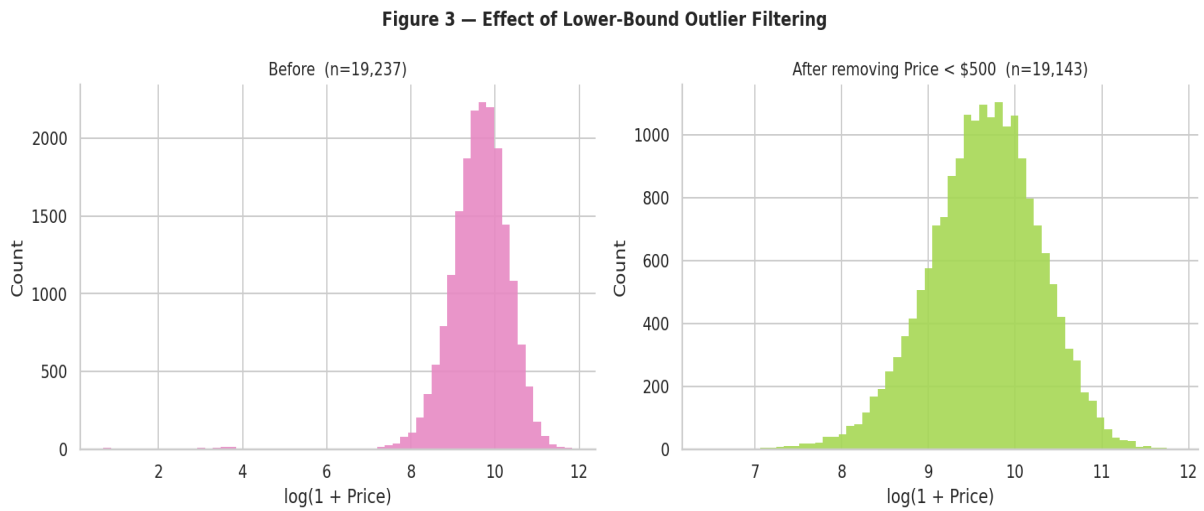


Figure 5 — Price distribution before and after filtering. High-value tail deliberately preserved.

## TRAIN / TEST SPLIT

The split is performed **before** any feature engineering or scaling. Fitting transformers on the full dataset first would allow test-set statistics to leak into training. The 80/20 split is **stratified by price decile** to ensure both sets share the same value distribution. Result: 15,316 training rows and 3,829 test rows.

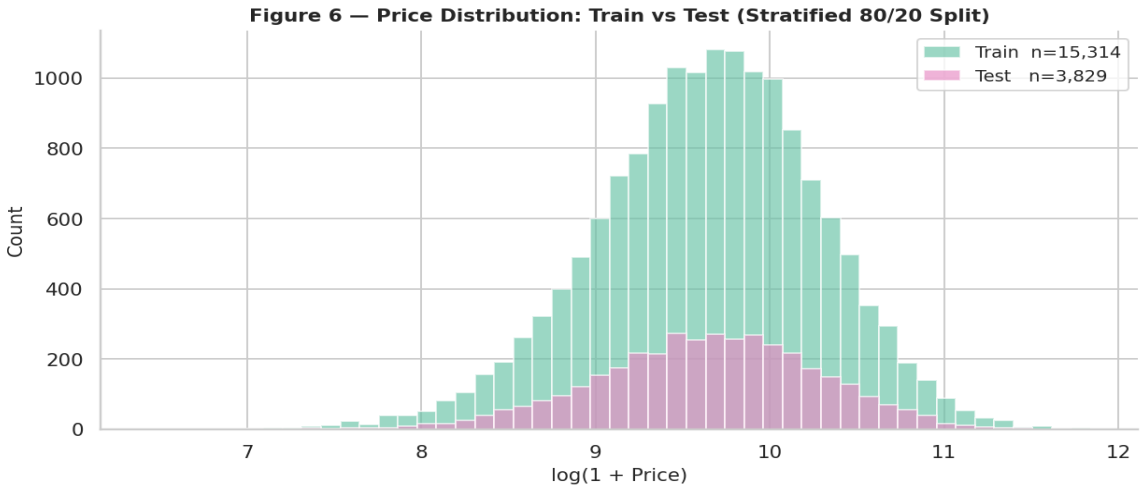


Figure 6 — Train vs test price distributions. Near-identical shapes confirm stratification worked.

## FEATURE ENGINEERING

Five hand-crafted features are derived from existing columns using domain knowledge about used car valuation. All computations happen after the split.

| Feature | Formula | Domain Reasoning |
|---|---|---|
| Car_Age | 2024 – Prod. year | Depreciation is time-driven |
| Km_Per_Year | Mileage ÷ (Car_Age + 1) | Normalises wear by ownership duration |
| Has_Turbo | 1 if raw string contains "Turbo" | Turbo engines command a price premium |
| Airbag_Tier | Binned: 0→0, 1-4→1, 5-8→2, 9+→3 | Safety tier, reduces outlier effect |
| Value_Score | (Eng×1000 – Mileage×0.01) ÷ (Age+1) | Composite: engine vs age and usage |

## ANALYSIS OF NEW FEATURES

Each new feature is analysed with the same methods used in Milestone 1: distribution histograms, regression plots, violin plots, KDE, and correlation matrix.

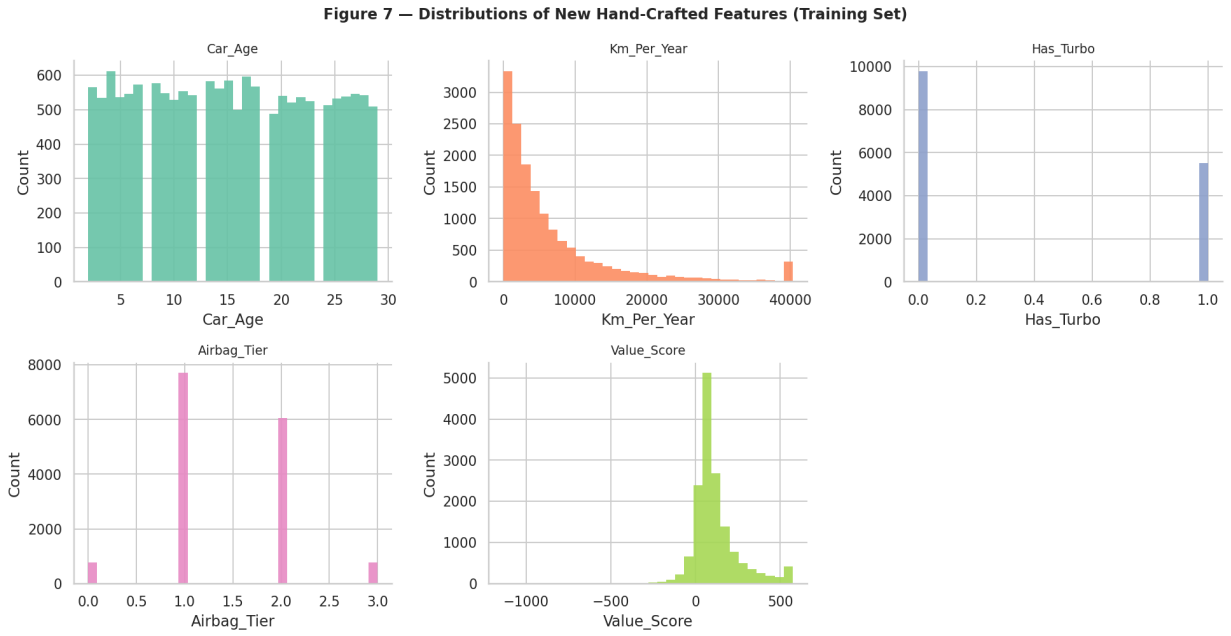


Figure 7 — Distributions of all five new features (training set).

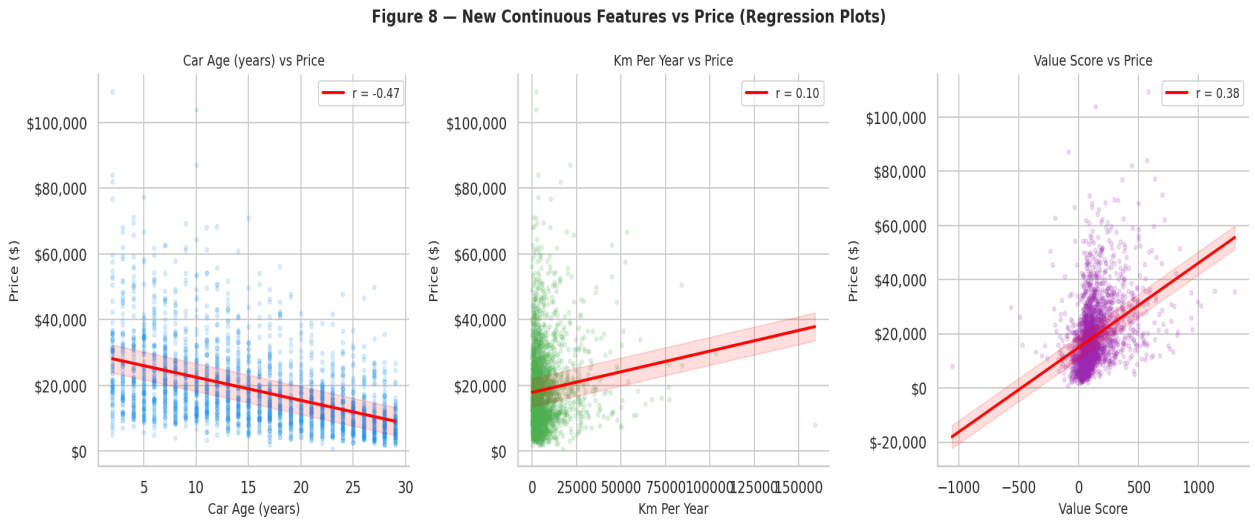


Figure 8 — Regression plots: Car Age, Km Per Year, Value Score vs Price with OLS trend lines.

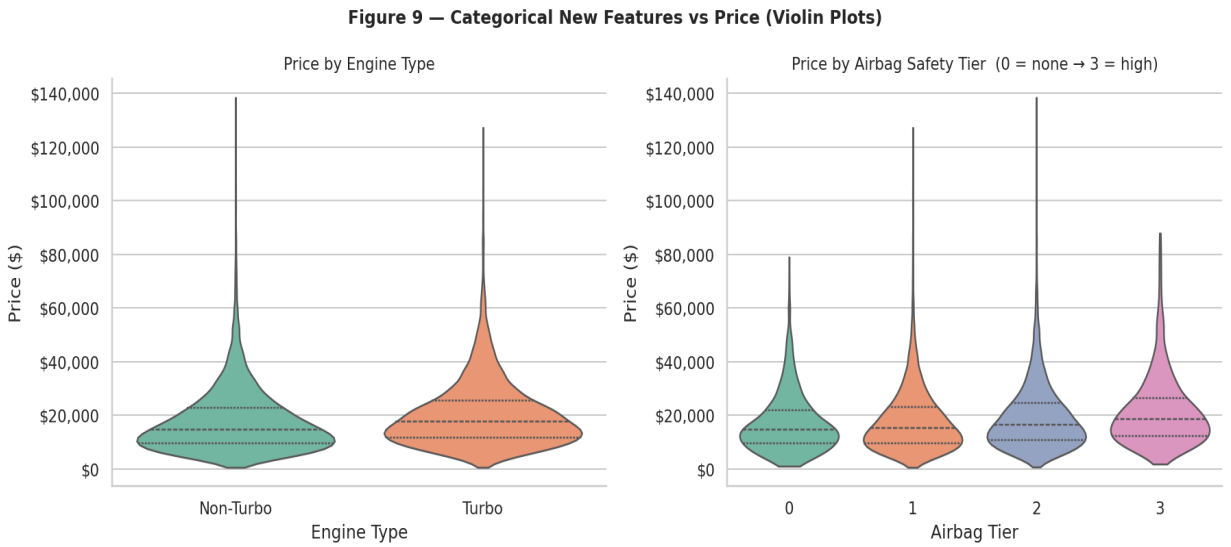


Figure 9 — Violin plots: turbo engines and higher airbag tiers correspond to higher median prices.

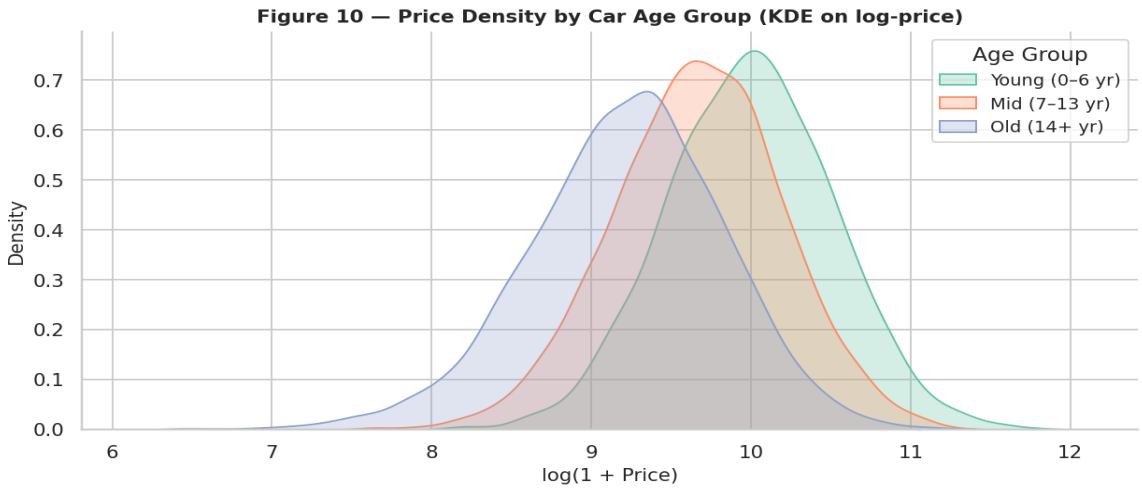


Figure 10 — KDE density by car age group. Distribution shifts leftward as age increases.

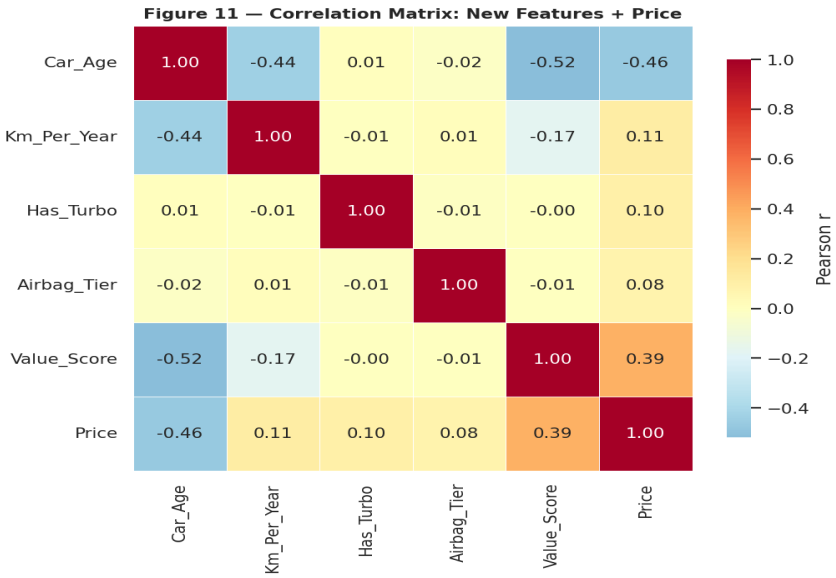


Figure 11 — Correlation matrix: new features + Price.

## ENCODING CATEGORICAL VARIABLES

**Target Encoding** (Manufacturer, Model, Color) — each category replaced by its mean Price in the training set. Ordinal encoding was rejected because Toyota=0, BMW=1 would impose a meaningless numeric order.

**One-Hot Encoding** (Fuel type, Gear box type, etc.) — binary indicator columns for low-cardinality features. No ordering imposed.

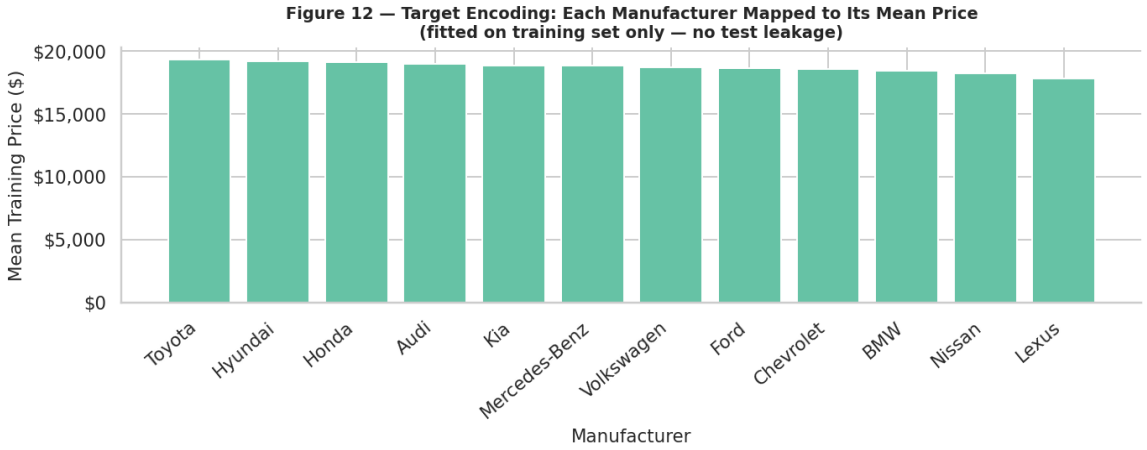


Figure 12 — Target encoding: each manufacturer mapped to its mean training-set price.

## SCALING VIA SKLEARN PIPELINE

A **Pipeline + ColumnTransformer** wraps all preprocessing. fit_transform() on training only, transform() on test — leakage is structurally impossible. **MinMaxScaler** maps all features to [0, 1]. A **SimpleImputer (median)** runs first inside the pipeline to handle any residual NaN values.

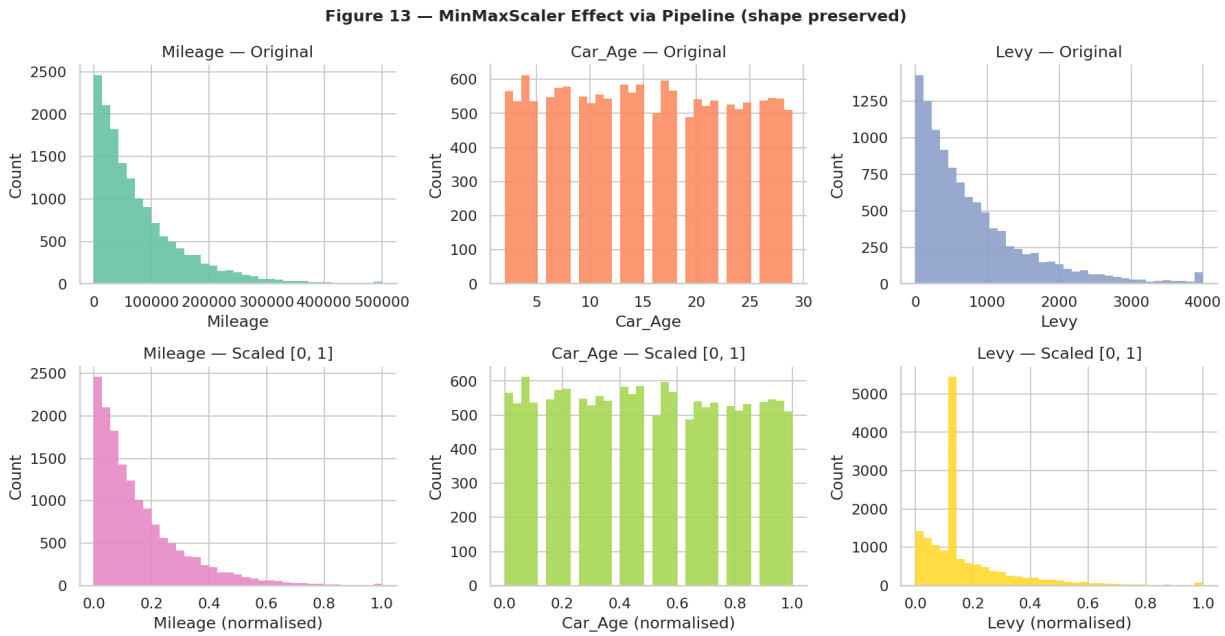


Figure 13 — MinMaxScaler via Pipeline. Distribution shape preserved; range normalised to [0, 1].

## FEATURE SELECTION

A **GradientBoostingRegressor** is trained and **Permutation Importance** computed — measuring how much error increases when each feature is randomly shuffled. Threshold: importance > 0.002. Reduces from 37 to **14 features**, dropping 23 low-signal columns.

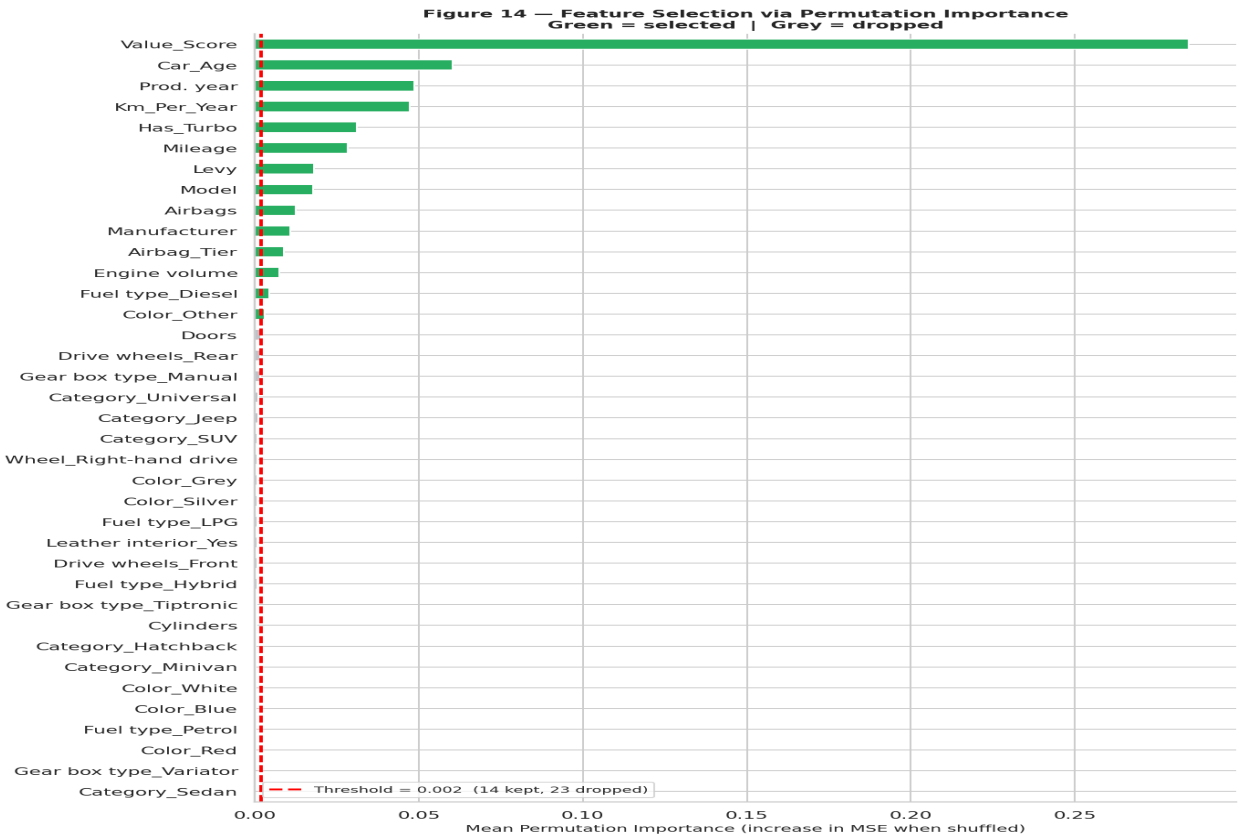


Figure 14 — Permutation Importance. 14 kept (green), 23 dropped (grey).

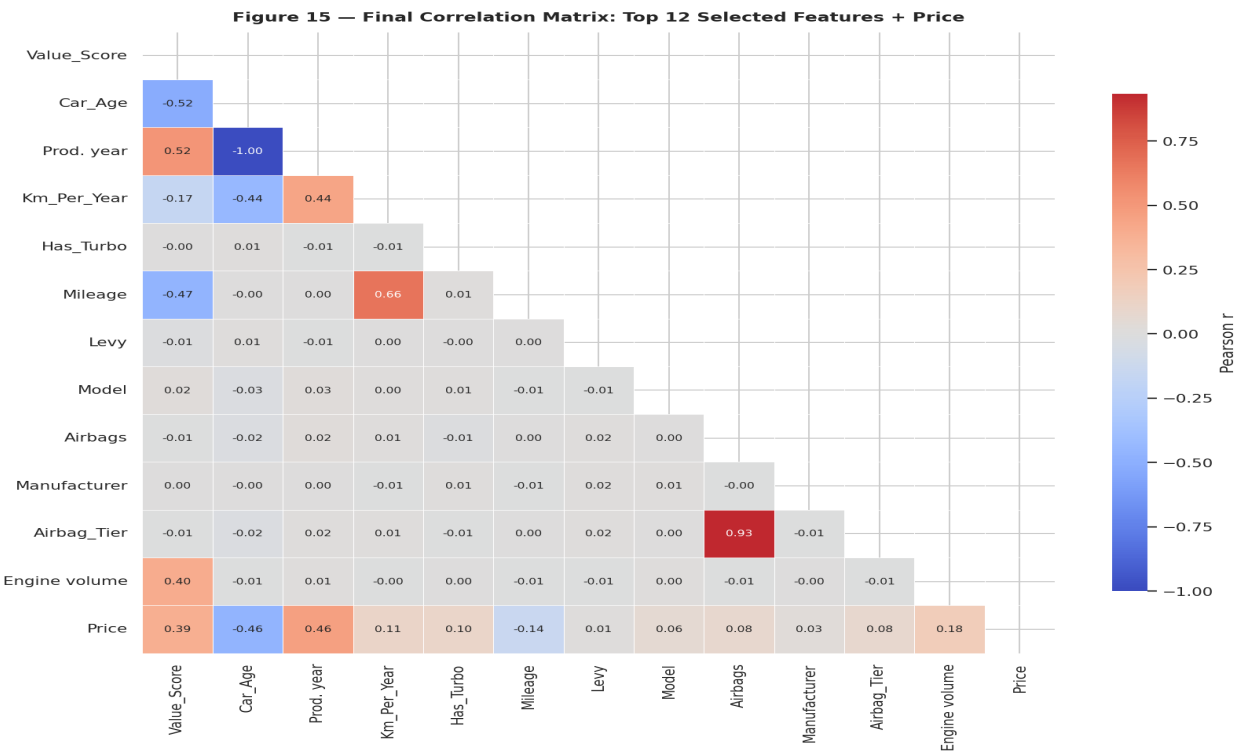


Figure 15 — Correlation matrix of final selected features + Price.

## SUMMARY

| Step | Method | Outcome |
|---|---|---|
| Type cleaning | Strip units, regex, pd.to_numeric | All columns typed correctly |
| Missing values | Median imputation; row drop for categoricals | Zero NaNs remain |
| Outlier filtering | Remove Price < $500 only | Luxury vehicles preserved |
| Train/test split | 80/20 stratified by price decile | Balanced target distribution |
| Feature engineering | 5 hand-crafted features (add_features()) | Domain-driven predictors added |
| Encoding | Target enc. (high-card) + OHE (low-card) | Fully numeric feature matrix |
| Scaling | SimpleImputer + MinMaxScaler via Pipeline | Leak-free [0,1] normalisation |
| Feature selection | Permutation Importance > 0.002 | 14 features retained |

## CONCLUSION

The raw car listings dataset has been fully transformed into a clean, encoded, scaled, and feature-selected matrix ready for model training in Milestone 3. Five meaningful features were engineered and validated against the target variable. A stratified 80/20 split ensures evaluation integrity, and the sklearn Pipeline guarantees that no test-set information influenced any fitted transformer.